



Databases and its Applications

Advanced NoSQL and Emerging Database trends

Pooja T S

Computer Applications

Databases and its Applications

Advanced NoSQL and Emerging Database trends

Vector Databases, AI/ML-Ready Storage

Pooja T S

Computer Applications



Databases and its Applications

Need for Vector Databases

► **Limitations of traditional databases**

Designed for exact matches and structured queries
Inefficient for similarity-based searches
Cannot represent semantic meaning effectively

► **Modern application requirements**

Search based on meaning, not exact values
Recommendation and personalization systems
AI-driven content understanding



What is a Vector Database?

► Core idea

Data is stored as numerical vectors

Each vector represents features of an object

► Key capability

Enables similarity search using distance metrics

Retrieves results that are semantically closest

► Contrast with relational storage

Focuses on similarity rather than equality



Vector Databases

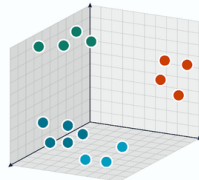
INPUT DATA (IMAGES)



VECTOR EMBEDDING

[[0.95, 0.49, 0.67, 0.75, 0.43, 0.00, 0.60, 0.13, 0.72, ...],
[0.73, 0.63, 0.80, 0.48, 0.55, 0.63, 0.86, 0.85, 0.06, ...],
[0.63, 0.71, 0.62, 0.07, 0.74, 0.30, 0.80, 0.08, 0.63, ...],
[0.49, 0.48, 0.38, 0.51, 0.45, 0.05, 0.10, 0.92, 0.29, ...],
[0.84, 0.38, 0.81, 0.00, 0.48, 0.61, 0.41, 0.85, 0.93, ...],
[0.24, 0.90, 0.69, 0.57, 0.43, 0.60, 0.37, 0.63, 0.28, ...],
[0.72, 0.33, 0.36, 0.98, 0.88, 0.05, 0.64, 0.91, 0.15, ...],
[0.89, 0.17, 0.44, 0.78, 0.33, 0.59, 0.29, 0.23, 0.22, ...],
[0.88, 0.75, 0.38, 0.77, 0.31, 0.98, 0.84, 0.79, 0.25, ...],
[0.11, 0.77, 0.30, 0.56, 0.62, 0.14, 0.89, 0.19, 0.62, ...],
[0.63, 0.41, 0.97, 0.73, 0.64, 0.41, 0.96, 0.36, 0.87, ...],
[0.51, 0.16, 0.26, 0.94, 0.53, 0.38, 0.39, 0.90, 0.55, ...],
[0.91, 0.99, 0.10, 0.87, 0.24, 0.91, 0.94, 0.36, 0.69, ...],
[0.89, 0.64, 0.14, 0.94, 0.51, 0.24, 0.61, 0.99, 0.11, ...],
[0.77, 0.65, 0.48, 0.94, 0.49, 1.00, 0.24, 0.10, 0.97, ...],
[0.58, 0.47, 0.01, 0.56, 0.59, 0.32, 0.79, 0.95, 0.40, ...]]

VECTOR DATABASE



altexsoft



How Vector Data is Generated?

► Feature extraction

Raw data (text, image, audio) is processed
Converted into numerical representations

► Vector characteristics

High-dimensional numerical arrays
Each dimension represents a learned feature

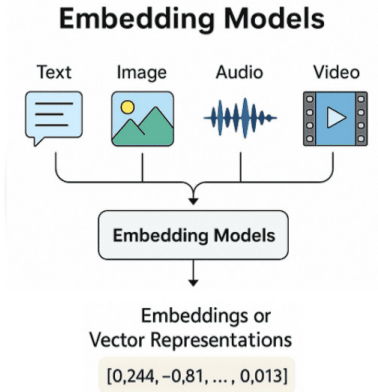
► Storage perspective

Database stores vectors along with metadata



Databases and its Applications

Vector Embeddings





Databases and its Applications

Similarity Search in Vector Databases

▶ **Search principle**

Finds vectors closest to a query vector
Uses distance or similarity measures

▶ **Common similarity goals**

Find items with similar meaning
Rank results by relevance

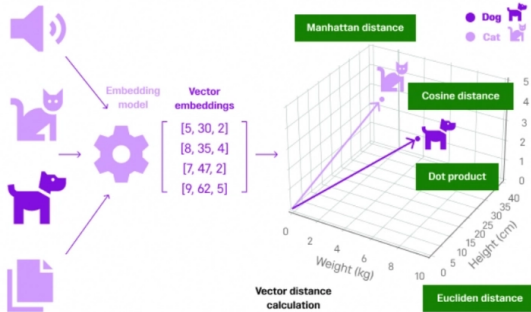
▶ **Outcome**

Results are approximate but semantically relevant



Databases and its Applications

Similarity Search





► Definition

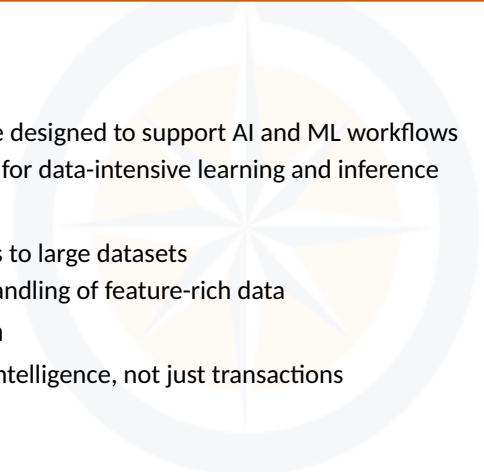
A database designed to support AI and ML workflows
Optimised for data-intensive learning and inference

► Core focus

Fast access to large datasets
Efficient handling of feature-rich data

► Key distinction

Supports intelligence, not just transactions





Why Traditional Databases Fall Short for AI/ML?

► Design mismatch

Built for OLTP-style workloads

Optimised for small, frequent updates

► AI/ML workload nature

Large batch reads

Feature extraction and model training

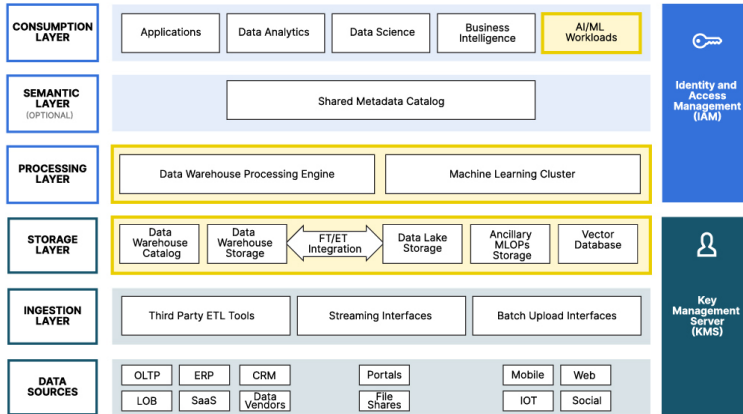
► Result

Performance bottlenecks in ML pipelines



Databases and its Applications

AI/ML Architecture





► **Nature of data**

Mostly unstructured or semi-structured
High dimensional and evolving

► **Examples**

Text embeddings
Image feature vectors

► **Storage implication**

Requires flexible, schema-light storage



Databases and its Applications

Architectural Features

► Scalability

Handles massive datasets

Scales horizontally with data growth

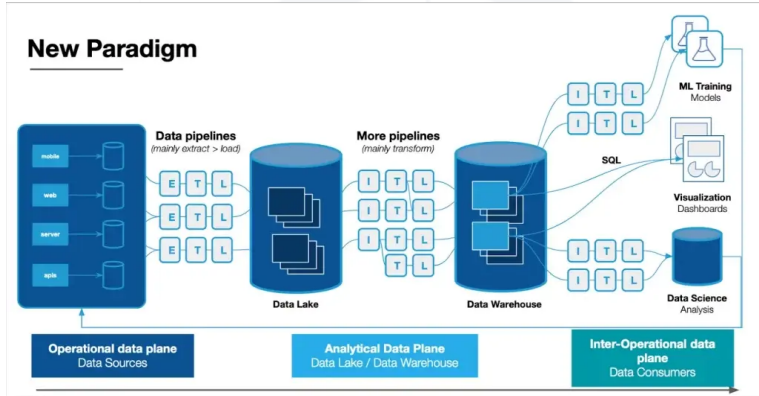
► Access patterns

Optimised for read-heavy operations

Supports parallel data access



Data Pipelines





Databases and its Applications

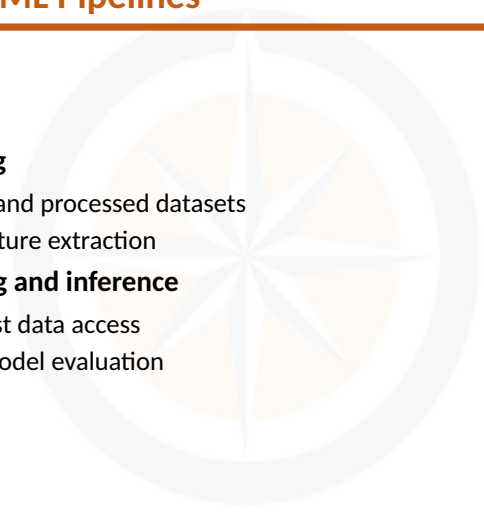
Role in ML Pipelines

► Before training

- Store raw and processed datasets
- Enable feature extraction

► During training and inference

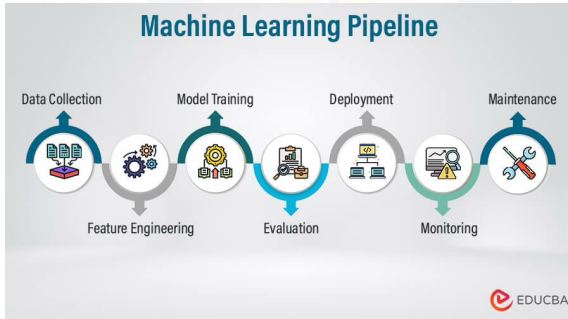
- Provide fast data access
- Support model evaluation





Databases and its Applications

Data Pipeline for AI/ML Systems





► **Why vectors matter**

ML models output numerical embeddings

These embeddings capture semantic meaning

► **Database role**

Store and index embeddings

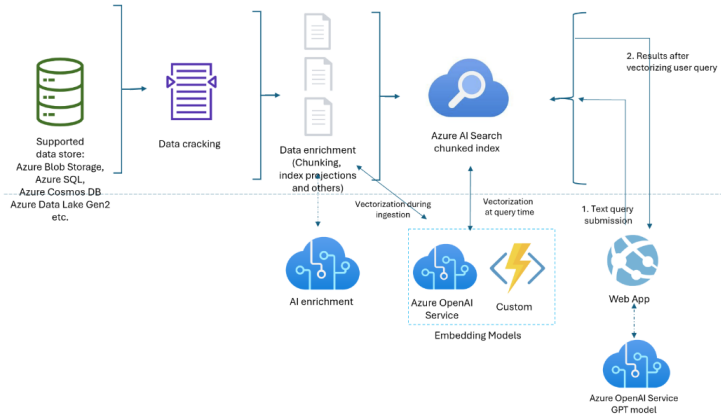
Enable similarity-based retrieval



Databases and its Applications

Azure AI- Example

Azure AI Search processing





PES
UNIVERSITY

CELEBRATING 50 YEARS

Pooja T S
Assistant Professor
Department of Computer Applications
poojats@pes.edu
080-26721983 Extn: 391